

MATC-Net: A Dual-Transformer Approach for WiFi-based Human Activity Recognition

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Abstract—WiFi-based Human Activity Recognition (HAR) offers a promising, privacy-preserving alternative to vision-based or wearable sensor systems by leveraging ubiquitous WLAN infrastructure. In this paper, we propose a novel dual-transformer encoder architecture, called MATC-Net (Multiscale convolutional Attention-based encoder for Temporal and Channel features with integrated fusion), which processes Channel State Information (CSI) in both temporal and channel dimensions to capture comprehensive spatiotemporal patterns. Our approach incorporates a multi-scale convolutional block within each encoder, enabling robust feature extraction over varying time scales, and an attention-based fusion mechanism to adaptively unify the separate representations. Experimental evaluations demonstrate high accuracy in classifying typical daily activities, highlighting the dominant role of the temporal encoder, yet underscoring the complementary benefits of channel-wise features. Crucially, we address real-world deployment challenges by introducing a transfer learning strategy that fine-tunes the model with minimal user data to accommodate different environments. This significantly boosts performance across diverse settings, illustrating that MATC-Net is both accurate and adaptable. Our findings provide valuable insights for designing future WiFi-based HAR systems and open avenues for broader applications in smart homes, healthcare, and beyond.

Index Terms—WiFi-based HAR, Transformers, Channel State Information, Multi-scale Convolution, Transfer Learning

I. INTRODUCTION

Human Activity Recognition (HAR) has gained significant attention due to its wide range of applications in smart homes, healthcare, security, and human-computer interaction. Traditional HAR methods, such as vision-based systems and wearable sensors, often suffer from privacy concerns, environmental constraints, and user inconvenience. To address these limitations, WiFi-based HAR has emerged as a promising solution, leveraging the omnipresent WiFi infrastructure to enable device-free, privacy-preserving activity recognition.

WiFi Channel State Information (CSI) captures fine-grained variations in signal propagation caused by human movements, making it highly suitable for activity recognition tasks [1]. Recent advancements in deep learning, particularly in convolutional and recurrent neural networks, have significantly improved HAR performance by automatically extracting meaningful features from CSI data [2]. However, these models often struggle to capture long-range dependencies and global feature correlations, which are essential for robust recognition, particularly in diverse real-world environments [3].

To address these limitations, transformer-based architectures have gained attention in HAR research for their ability to model complex dependencies using self-attention mechanisms

[4]. In this work, we explore a novel HAR framework leveraging a dual transformer encoder architecture. The first encoder focuses on extracting temporal features, while the second processes channel-wise dependencies in WiFi CSI data. These outputs are fused via an attention-based mechanism to create a unified representation that captures both spatial and temporal aspects of the signal.

Our work first investigates the performance of the complete model, followed by a detailed analysis of the individual contributions of the two encoders. Additionally, we aim to provide a visual interpretation of the outputs generated by these transformer encoders, offering insights into their behavior and feature representations. Additionally, we study transfer learning mechanisms to fine-tune the model on data from new environments, enabling it to adapt to varying conditions effectively. This approach aims to reduce the reliance on large labeled datasets by calibrating the model with a minimal amount of user-provided data during deployment.

This paper is structured as follows: Section II reviews related work on WiFi-based HAR using transformer architecture e other Deep learning models. Section III present the preprocessing data format and the preprocessing techniques used. Section V presents the proposed system model and methodology. Section VI details the experimental setup and results, including the evaluation of different encoder designs and transfer learning mechanisms. Finally, Section IX concludes the paper with directions for future research.

II. RELATED WORK

Recent advancements in WiFi-based Human Activity Recognition (HAR) have demonstrated the potential of using wireless signal perturbations as a non-invasive alternative to traditional methods. These methods rely on the analysis of Channel State Information (CSI), which captures the fine-grained impact of human movement on wireless signals. CSI provides a richer representation compared to simpler metrics like Received Signal Strength Indicator (RSSI), enabling more accurate and robust activity recognition.

A subset of research in CSI-based HAR, particularly the works that inspired our approach, can be broadly divided into two main directions: temporal dependency modeling and channel-wise spatial analysis. These studies, like ours, focus exclusively on leveraging the amplitude of CSI, rather than its phase information. Temporal modeling focuses on capturing the sequential dynamics of activities, while channel-wise analysis emphasizes spatial variations across subcarriers

and antennas. Both perspectives contribute complementary insights into activity recognition.

On the temporal side, recurrent neural networks (RNNs) and their variants, such as Long Short-Term Memory (LSTM) networks and Bidirectional Gated Recurrent Units (BGRUs), have been extensively used to capture the sequential nature of human activities [1]. More recently, transformer-based architectures, such as Vision Transformers (ViTs), have been successfully adapted for HAR tasks, demonstrating their capability to model long-range dependencies explicitly and efficiently [3].

Channel-wise analysis, on the other hand, has leveraged convolutional neural networks (CNNs) to extract spatial features across subcarriers. Attention mechanisms, such as the Convolutional Block Attention Module (CBAM), have further enhanced feature extraction by focusing on the most relevant spatial regions [1]. These methods highlight the importance of spatial information in distinguishing between activities with subtle signal variations.

Hybrid models that combine temporal and spatial analyses have shown significant promise in improving HAR performance. The AF-ACT framework [1] integrates CNNs with CBAM for channel-wise analysis and BGRUs with self-attention for temporal feature modeling, achieving state-of-the-art results. Similarly, Multi-scale Convolution Transformers (MSCT) [4] and Two-Stream Convolution Augmented Transformers (THAT) [2] employ multi-scale convolutional blocks alongside self-attention mechanisms to capture both local and global features. These approaches demonstrate the value of fusing complementary perspectives for complex activity recognition tasks.

Our work is directly inspired by THAT [2], which effectively combines convolutional and transformer-based layers to address the dual challenges of temporal and spatial feature extraction. However, while THAT treats these analyses separately, our model, MATC-Net, introduces a unified dual-transformer architecture that explicitly integrates temporal and channel-wise analyses. By employing an attention-based fusion mechanism, MATC-Net bridges these two dimensions, enabling the model to capture both long-range dependencies and spatial variations in a cohesive framework.

Transformer architectures, with their self-attention mechanisms, offer unique advantages in modeling relationships across both time and space. Unlike RNNs, transformers process sequences in parallel, reducing computational overhead while capturing complex dependencies [5]. This characteristic has driven their adoption in HAR tasks, where CSI data is treated as multivariate time-series data. Moreover, the adaptability of transformers allows them to generalize effectively across different environments, making them suitable for real-world deployments [3].

In this work, we expand on the transformer paradigm by designing a multi-scale convolutional block within each transformer encoder. This enhances the model's ability to capture features over varying temporal ranges, which is particularly important for recognizing activities with diverse durations.

Furthermore, the attention-based fusion mechanism enables the integration of temporal and channel-wise representations, yielding a more robust and adaptable solution for WiFi-based HAR. Our contributions not only advance the state of the art but also provide a scalable framework that can be extended to broader applications in smart homes, healthcare, and other domains.

III. PROCESSING PIPELINE

In this section, we present a high-level description of the processing pipeline employed in our work. The pipeline begins with data acquisition using the Nexmon CSI tool. This step involves capturing Channel State Information (CSI) either during the execution of a specific activity within a room or in an empty room to serve as a baseline. Once the data is collected, it undergoes a preprocessing phase, which includes noise reduction techniques. Specifically, we employ a Hampel filter and wavelet transform denoising to effectively remove noise from the signals. Following the denoising phase, the processed signals are segmented into shorter temporal sequences, making them suitable for further analysis by our model.

After preprocessing, the pipeline proceeds to the model phase, where the MATC-Net is utilized. The data is processed by two transformer encoder towers. The first tower analyzes the temporal dimension of the sequence, preceded by a positional embedding step to encode temporal information appropriately. The second tower examines the inter-channel correlations across subcarriers. Each of these towers is equipped with a Multi-Scale Convolutional Block to capture features at various scales.

The outputs from the two encoder towers are then combined using an Attention Fusion Block. This block aggregates the temporal and channel-based features into a single feature vector, which is subsequently fed into the final classification layer. This comprehensive pipeline effectively integrates temporal and spatial features, ensuring accurate classification results in our experiments.

IV. SIGNALS AND FEATURES

In this section, we detail the signals used in our study and the methodologies employed for feature extraction. We first describe the measurement setup and the format of the input data. Subsequently, we elaborate on the signal preprocessing steps and the extraction of feature vectors from the pre-processed signals, including the segmentation strategy adopted.

A. CSI data

Wi-Fi systems employ Orthogonal Frequency Division Multiplexing (OFDM), transmitting user data over K partially overlapping and orthogonal subcarriers. At the OFDM receiver, channel parameters, amplitudes and phases are continuously estimated for all subcarriers, resulting in the CFR, a complex matrix that characterizes the channel for each subcarrier and receiving antenna. For each pair of transmit

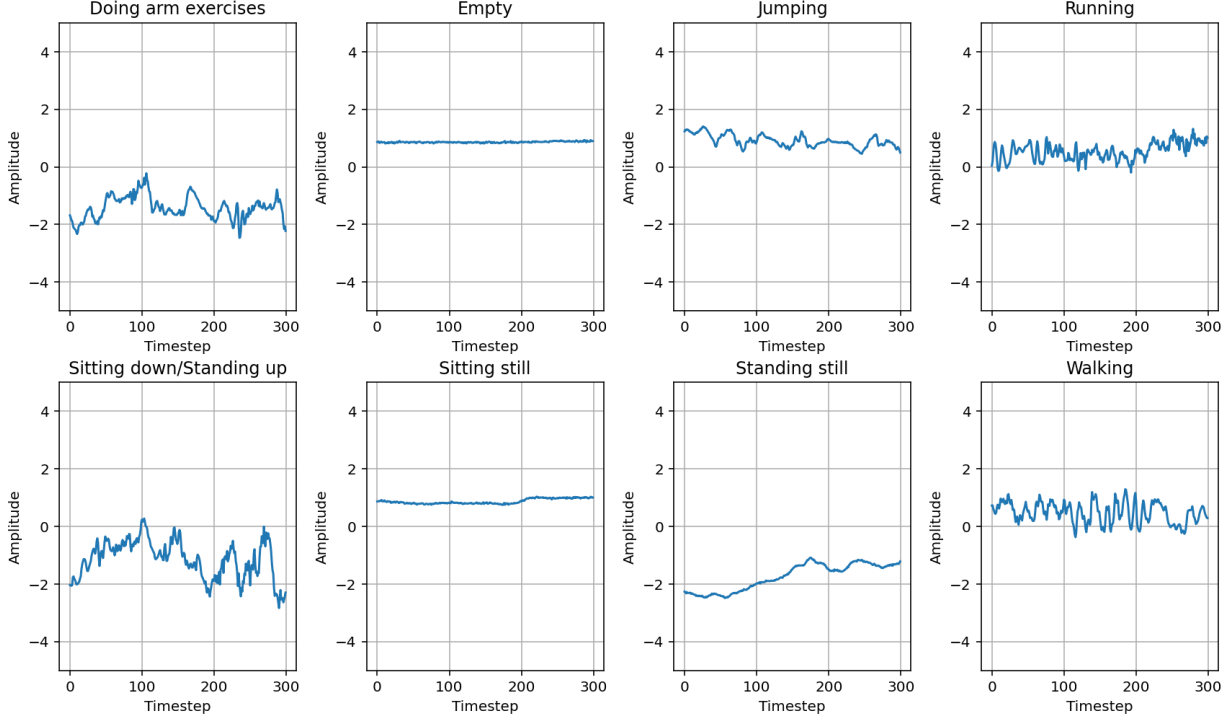


Fig. 1: Amplitude over time for a single subcarrier and antenna across different activities

and receive antennas, the CFR consists of complex numbers $A_k e^{j\phi_k}$, specifying the attenuation A_k and phase shift ϕ_k for each subcarrier $k \in -\frac{K}{2}, \dots, \frac{K}{2} - 1$. Considering a multipath propagation channel with P paths, the CFR estimated at packet n is expressed as:

$$H_k(n) = A_k(n) e^{j\phi_k(n)} = \sum_{p=0}^{P-1} A_p(n) e^{j2\pi(f_c + \frac{k}{T})\tau_p(n)}, \quad (1)$$

where f_c is the main carrier frequency, $T = \frac{1}{\Delta f}$ is the OFDM symbol duration with subcarrier spacing Δf , and $A_p(n)$ and $\tau_p(n)$ represent the attenuation and delay associated with path p at packet n . The collected CFR data may deviate from the theoretical model due to hardware artifacts, introducing an undesired phase offset $\phi_{\text{offs},k}$:

$$\tilde{H}k(n) = H_k(n) e^{j\phi_{\text{offs},k}}. \quad (2)$$

We proceed to describe the steps implemented to process the CFR matrices extracted by the Nexmon CSI tool. A new CFR matrix is retrieved for each received packet, leading to variable intervals between subsequent CFR acquisitions because of network traffic patterns. For the purposes of our analysis, we assume that a new sample is available at fixed intervals of T_c seconds to simplify the exposition. It is noteworthy that in real Wi-Fi networks, the existing traffic exchanged between access points and user devices can be captured and exploited for HAR applications. When such traffic is

insufficient or not available, two Wi-Fi access points can be configured to recreate the setup. This involves using a simple network controller to transmit packets at regular intervals, while a monitoring device collects the CFR data from these transmissions. In our experimental setup, we employ a Wi-Fi monitoring device running the Nexmon CSI tool to extract CFR matrices from the received packets. The CFR data is organized into time sequences, with each sample represented by a complex matrix of dimensions $N_{\text{sub}} \times N_{\text{ant}}$, where N_{sub} is the number of subcarriers and N_{ant} is the number of receiving antennas.

B. Signal Pre-Processing

The signal pre-processing pipeline is structured into three main stages: outliers removal with Hampel filter, denoising of raw WiFi CSI data using a wavelet transform, and temporal segmentation followed by reshaping. This reorganization ensures clarity and aligns with the actual sequence of operations performed.

a) Hampel Filtering: To remove outliers, a Hampel filter is applied. This filter identifies anomalies by comparing values with the median of their local neighborhood. Anomalous values that deviate beyond a specified threshold are replaced by the median, ensuring robust outlier mitigation.

b) Wavelet Transform Denoising: This approach effectively reduces random noise while preserving essential signal

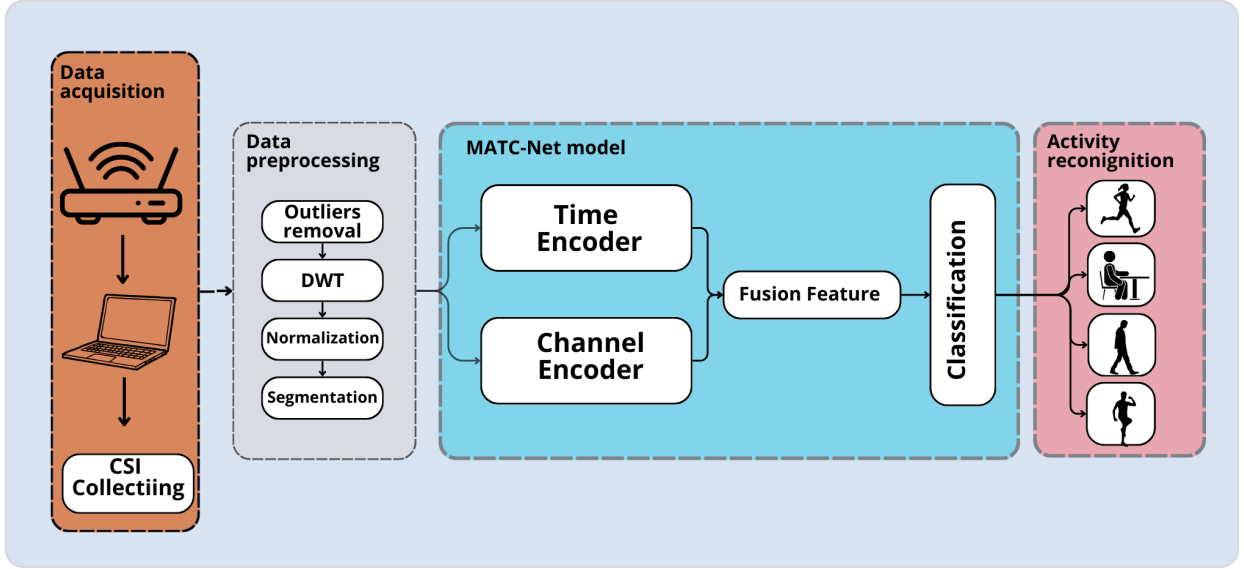


Fig. 2: Overall architecture of the proposed MATC-Net approach.

structures, aiding human activity recognition [6]. Given the raw CSI data, expressed as:

$$H_i = [H_{1,i}e^{j\angle H_{1,i}}, H_{2,i}e^{j\angle H_{2,i}}, \dots, H_{m,i}e^{j\angle H_{m,i}}],$$

a multi-resolution wavelet decomposition is performed using the pyramid algorithm, which can be expressed as:

$$H_{\kappa,i} = \sum_{k \in \mathbb{Z}} \langle H_{\kappa,i}, \phi_{j,k}(\kappa) \rangle \phi_{j,k}(\kappa) + \sum_{j \leq j' \leq J} \sum_{k \in \mathbb{Z}} \langle H_{\kappa,i}, \psi_{j',k}(\kappa) \rangle \psi_{j',k}(\kappa), \quad (3)$$

where $\phi_{j,k}(\kappa)$ and $\psi_{j,k}(\kappa)$ are the scaling and wavelet functions, respectively. The approximation ($c_{j,k}$) and detail ($d_{j,k}$) coefficients are computed as:

$$c_{j,k} = \langle H_{\kappa,i}, \phi_{j,k}(\kappa) \rangle, \quad (4)$$

$$d_{j,k} = \langle H_{\kappa,i}, \psi_{j,k}(\kappa) \rangle. \quad (5)$$

Noise reduction is achieved through soft thresholding of the detail coefficients. The threshold T is dynamically calculated based on the variability of the detail coefficients:

$$T \propto \sigma \sqrt{\ln(n)}, \quad (6)$$

where σ is a robust estimate of noise variability and n is the length of the signal.

The soft thresholding function is defined as:

$$S(d_{j,k}) = \begin{cases} d_{j,k} - T, & d_{j,k} > T, \\ d_{j,k} + T, & d_{j,k} < -T, \\ 0, & |d_{j,k}| \leq T, \end{cases} \quad (7)$$

where $d_{j,k}$ are the detail coefficients. This step suppresses small coefficients, corresponding to noise, while preserving the significant components of the signal.

The denoised signal is reconstructed using the inverse

wavelet transform:

$$\hat{H}_{\kappa,i} = \sum_{k \in \mathbb{Z}} c_{J,k} \phi_{J,k}(\kappa) + \sum_{j=-\infty}^J \sum_{k \in \mathbb{Z}} \hat{d}_{j,k} \psi_{j,k}(\kappa), \quad (8)$$

where $\hat{H}_i = [\hat{H}_{1,i}e^{j\angle H_{1,i}}, \hat{H}_{2,i}e^{j\angle H_{2,i}}, \dots, \hat{H}_{m,i}e^{j\angle H_{m,i}}]$ represents the denoised WiFi CSI data.

c) *Amplitude Extraction and Hampel Filtering*: For our analysis, we utilized the module of the CSI, specifically focusing on the amplitude of the CSI rather than the phase. This decision was made to avoid the need for phase sanitization, which can introduce additional complexities and uncertainties. Instead, the amplitude, denoted as $|\hat{H}(n)|$, provides a robust basis for our signal processing pipeline. From this point onward, we will refer to the amplitude of the CSI simply as H .

d) *Segmentation and Reshaping*: The CSI amplitude is segmented into smaller overlapping temporal windows of W samples. The denoised and segmented amplitude matrix, originally in the form $H \in \mathbb{R}^{T \times S \times A}$, where T , S , and A represent the number of time samples, subcarriers, and antennas, respectively, is then reshaped into $\mathbf{H}' \in \mathbb{R}^{T \times d_{in}}$, with $d_{in} = S \cdot A$. This transformation flattens the subcarrier and antenna dimensions into a single feature dimension, preparing the data for further analysis.

V. LEARNING FRAMEWORK

The MATC-Net is composed of two encoder towers: one operates along the temporal dimension, while the other processes the feature dimension, specifically the subcarriers. The outputs of these two encoders are then combined through an Attention Fusion Block and subsequently passed to the final classification layers.

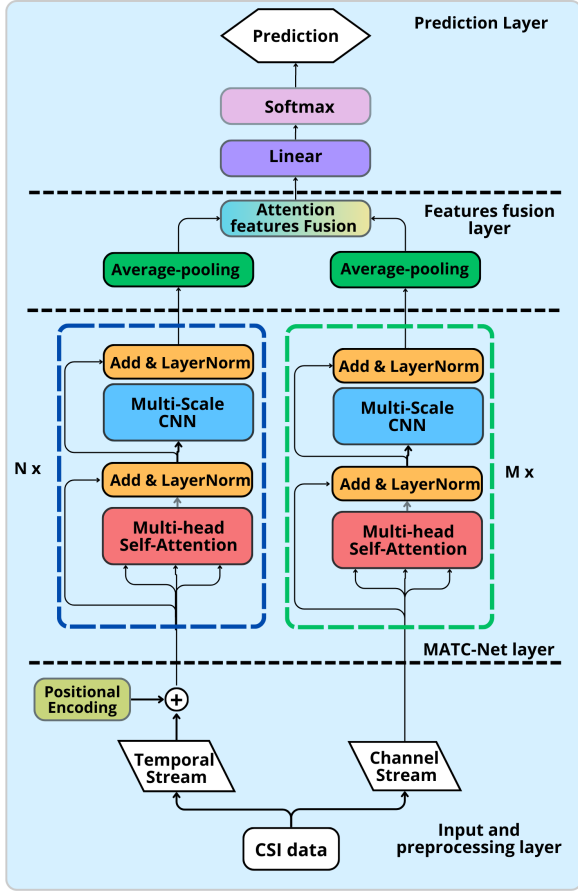


Fig. 3: Architecture of the MATC-Net.

A. Encoder Structure

The input data first passes through an embedding layer. This embedding layer is implemented using a feedforward layer, where the output is projected through a linear transformation to match the embedding dimension of the features. For the temporal encoder, the input $\mathbf{H}' \in \mathbb{R}^{T \times d_{in}}$ is reshaped and processed. The channel encoder operates analogously, with an initial transposition to facilitate analysis over the channel dimension. The input for this encoder will be $\mathbf{H}'' \in \mathbb{R}^{d_{in} \times T}$.

For the Time Encoder the linear layer applies a matrix multiplication without an activation function, defined as:

$$\mathbf{Y} = \mathbf{H}'\mathbf{W} + \mathbf{b}, \quad (9)$$

where $\mathbf{H}' \in \mathbb{R}^{T \times d_{in}}$ is the input, $\mathbf{W} \in \mathbb{R}^{d_{in} \times d_{emb}}$ is the weight matrix, $\mathbf{b} \in \mathbb{R}^{d_{emb}}$ is the bias, and $\mathbf{Y} \in \mathbb{R}^{T \times d_{emb}}$ is the output. Here, d_{in} and d_{emb} denote the input and output dimensions, respectively.

For the the Channel Encoder, the input $\mathbf{H}'' \in \mathbb{R}^{d_{in} \times T}$ undergoes a similar process. The embedded output $\mathbf{Z}$ is defined as:

$$\mathbf{Z} = \mathbf{H}''\mathbf{W}_C + \mathbf{b}_C, \quad (10)$$

where $\mathbf{W}_C \in \mathbb{R}^{T \times d_{emb}}$ and $\mathbf{b}_C \in \mathbb{R}^{d_{emb}}$. As a result, $\mathbf{Z} \in \mathbb{R}^{d_{in} \times d_{emb}}$. This projection aligns the features into the embedding space used for subsequent analysis.

For the Temporal Encoder, positional embedding is applied after the linear transformation to account for the temporal progression of the data, as introduced in [5]. After embedding, the data passes through a multi-head self-attention layer. Both the channel and temporal encoders are stacked in multiple layers, allowing the model to progressively refine the feature representation at each stage.

The detailed process of passing data through the Temporal Encoder will be discussed in subsequent sections V-B and V-C, noting that the Channel Encoder follows an analogous approach without the positional encoding since we are not interested in studying the temporal dependency. Both encoder outputs are then unified in the Attention Fusion Block, which will be described in Section V-D.

B. Multi-Head Self-Attention Module

We employ a multi-head self-attention mechanism to generate hidden representations from inputs. This mechanism integrates information across the entire sequence, effectively addressing the long-range dependency issues of RNNs and LSTMs.

The self-attention mechanism computes a weighted sum of the values, where the weight assigned to each value is determined by the dot product of the query and the corresponding key. Formally, it is defined as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} \right) \mathbf{V}, \quad (11)$$

where $\mathbf{Q} \in \mathbb{R}^{T \times d_k}$, $\mathbf{K} \in \mathbb{R}^{T \times d_k}$, and $\mathbf{V} \in \mathbb{R}^{T \times d_v}$ represent the queries, keys, and values, respectively. These are generated by linear projections of the input $\mathbf{Y} \in \mathbb{R}^{T \times d_{emb}}$ as:

$$\mathbf{Q} = \mathbf{Y}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{Y}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{Y}\mathbf{W}_V, \quad (12)$$

where $\mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{d_{emb} \times d_k}$, $\mathbf{W}_V \in \mathbb{R}^{d_{emb} \times d_v}$ are learnable projection matrices.

To jointly attend to information from different representation subspaces, the queries, keys, and values are independently projected multiple times (across h heads) with different projection parameters. The results are concatenated and projected again to yield the final output:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = [\text{head}_1; \dots; \text{head}_h]\mathbf{W}_O, \quad (13)$$

where $\text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_Q^i, \mathbf{K}\mathbf{W}_K^i, \mathbf{V}\mathbf{W}_V^i)$, and $\mathbf{W}_O \in \mathbb{R}^{h \cdot d_v \times d_o}$ is the final projection matrix. The final output of the attention mechanism, therefore, has a dimension of $\mathbb{R}^{T \times d_o}$, where d_o is the desired output size defined by the projection matrix $\mathbf{W}_O$. This dimension depends on the number of attention heads h and the size of each head's output d_v , as $d_o = h \cdot d_v$. To facilitate residual connections, all sub-layers, as well as the input layer, share the same dimension $d_{emb} = h \cdot d_v = d_o$.

C. Multi-Scale Convolutional Block

After the multi-head attention block, the output has shape $\mathbb{R}^{T \times d_o}$. It is then passed through a Multi-Scale Convolutional Block (MSCB) [2] instead of the feedforward layer used in

the original transformer architecture [5]. The MSCB captures features over varying temporal ranges to better represent human activity patterns.

Given a set of kernel sizes $\mathcal{J} = \{j_1, j_2, \dots, j_{|\mathcal{J}|}\}$, each kernel size corresponds to a specific temporal scale. Let $\mathbf{X} \in \mathbb{R}^{T \times d_o}$ be the input to the MSCB. For a kernel size j_i , the convolutional output $\mathbf{P}_i \in \mathbb{R}^{T \times d_o}$ is computed as:

$$\mathbf{P}_i = \text{ReLU}(\text{Dropout}(\text{BN}(\text{Conv}(\mathbf{W}_{j_i}; \mathbf{X})))), \quad (14)$$

where $\text{Conv}(\mathbf{W}_{j_i}; \mathbf{X})$ applies a 1D convolution operation with weights $\mathbf{W}_{j_i} \in \mathbb{R}^{d_o \times j_i \times d_o}$. The result is batch-normalized (BN), followed by dropout and a ReLU activation.

To adaptively combine the multi-scale features, we use an adaptive scale attention mechanism. The ensembled feature map $\mathbf{P}_{\text{ens}} \in \mathbb{R}^{T \times d_o}$ is computed as:

$$\mathbf{P}_{\text{ens}} = \sum_i^{|\mathcal{J}|} \alpha_i \mathbf{P}_i, \quad (15)$$

where α_i is the attention score for scale j_i . The attention scores $\alpha = [\alpha_1, \alpha_2, \dots, \alpha_{|\mathcal{J}|}]$ are computed as:

$$\alpha = \text{softmax}(\text{FFN}(\mathbf{P})), \quad (16)$$

with the feedforward network defined as:

$$\text{FFN}(\mathbf{P}_i) = \text{ReLU}(\mathbf{P}_i \mathbf{W}_1 + \mathbf{b}_1) \mathbf{W}_2 + \mathbf{b}_2, \quad (17)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_o \times d_h}$ and $\mathbf{W}_2 \in \mathbb{R}^{d_h \times 1}$ are learnable parameters, and $\mathbf{b}_1, \mathbf{b}_2$ are biases. The described process is illustrated in Figures 4 and ?? . Figure ?? depicts the convolution operation performed using multiple filters, while Figure ?? shows the integration of the extracted feature maps into a unified representation.

The MSCB thus enables the model to capture temporal features across multiple scales, ensuring robustness and adaptability in human activity recognition tasks.

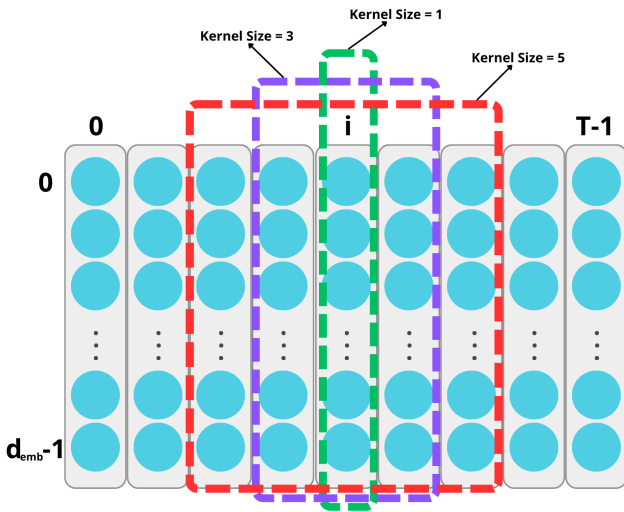


Fig. 4: Convolutions with kernel size 1, 3, and 5 in Time Encoder

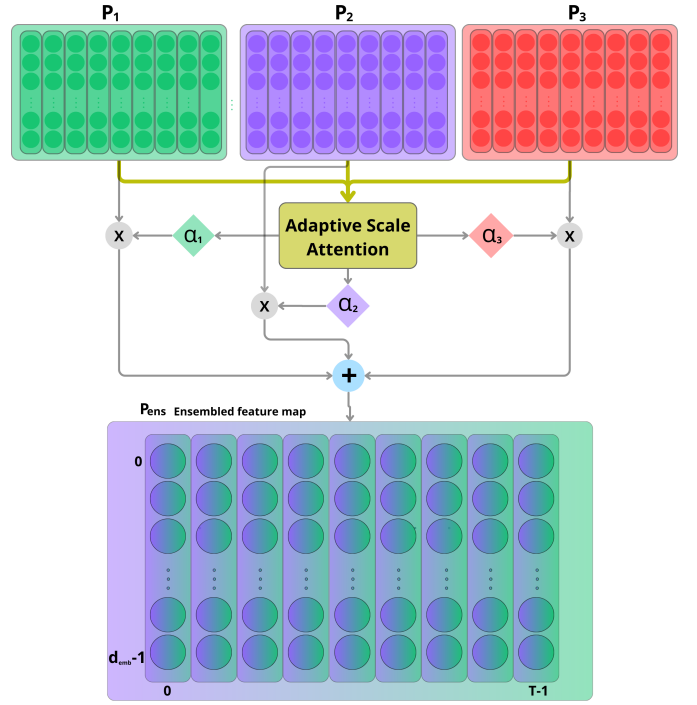


Fig. 5: Generation of the ensembled feature map $\mathbf{P}_{\text{ens}}$ from $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$, each with dimension $\mathbb{R}^{T \times d_o}$ in Time Encoder.

D. Attention features Fusion Block

After the outputs have passed through their respective encoder towers, two outputs are obtained: the temporal encoder output $\mathbf{O}_T \in \mathbb{R}^{T \times d_{\text{emb}}}$ and the transposed channel encoder output $\mathbf{O}_C \in \mathbb{R}^{d_{\text{in}} \times d_{\text{emb}}}$. Notably, the shapes of these outputs remain consistent with their respective encoder designs.

To reduce the dimensionality of these outputs while retaining their essential features, a Global Average Pooling operation is applied to both $\mathbf{O}_T$ and $\mathbf{O}_C$:

$$\mathbf{G}_T = \frac{1}{T} \sum_{t=1}^T \mathbf{O}_T^{(t)}, \quad \mathbf{G}_C = \frac{1}{d_{\text{in}}} \sum_{i=1}^{d_{\text{in}}} \mathbf{O}_C^{(i)}. \quad (18)$$

where $\mathbf{G}_T \in \mathbb{R}^{d_{\text{emb}}}$ and $\mathbf{G}_C \in \mathbb{R}^{d_{\text{emb}}}$ are the globally pooled feature vectors for the temporal and channel encoders, respectively.

1) *Projection onto a Unified Dimension:* The globally pooled feature vectors $\mathbf{G}_T$ and $\mathbf{G}_C$ are projected onto a common embedding dimension d_f using separate dense layers with ReLU activation:

$$\mathbf{P}_T = \text{ReLU}(\mathbf{G}_T \mathbf{W}_T + \mathbf{b}_T), \quad \mathbf{P}_C = \text{ReLU}(\mathbf{G}_C \mathbf{W}_C + \mathbf{b}_C), \quad (19)$$

where $\mathbf{W}_T, \mathbf{W}_C \in \mathbb{R}^{d_{\text{emb}} \times d_f}$ are the learnable projection weights, and $\mathbf{b}_T, \mathbf{b}_C \in \mathbb{R}^{d_f}$ are biases. This step ensures both feature vectors are aligned in the same embedding space.

2) *Concatenation for Attention Computation:* The projected vectors $\mathbf{P}_T$ and $\mathbf{P}_C$ are concatenated along a new axis to form a tensor:

$$\mathbf{C} = \text{stack}(\mathbf{P}_T, \mathbf{P}_C) \in \mathbb{R}^{2 \times d_f}. \quad (20)$$

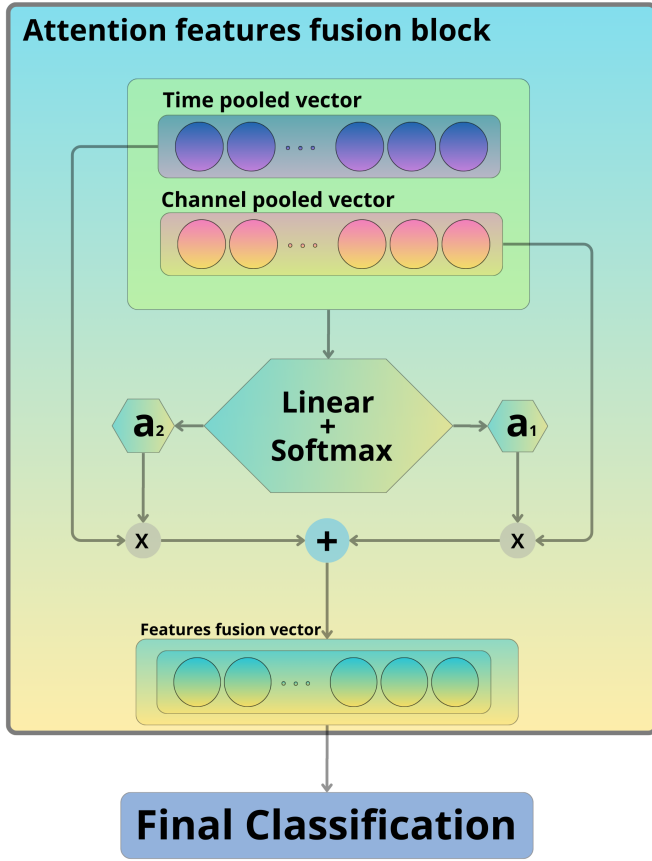


Fig. 6: Illustration of Attention features Fusion Block

3) *Attention-Based Fusion*: Attention scores are computed using a dense layer followed by a softmax function:

$$\mathbf{a} = \text{softmax}(\mathbf{C}\mathbf{W}_a + \mathbf{b}_a), \quad (21)$$

where $\mathbf{W}_a \in \mathbb{R}^{d_f \times 1}$ and $\mathbf{b}_a \in \mathbb{R}$ are learnable parameters. The resulting attention weights $\mathbf{a} \in \mathbb{R}^{2 \times 1}$ indicate the importance of each representation.

Weighted Fusion: The final unified representation $\mathbf{F} \in \mathbb{R}^{d_f \times 1}$ is computed as the weighted sum of the projected vectors:

$$\mathbf{F} = \sum_{i=1}^2 a_i \mathbf{C}_i, \quad (22)$$

where a_i represents the attention weight for the i -th representation, and $\mathbf{C}_i$ is the corresponding vector.

This attention fusion mechanism ensures that the model adaptively integrates temporal and channel-based features, leveraging their complementary strengths for robust activity recognition.

E. Final Classification Layers

Subsequently, the feature-fused vector $\mathbf{F}$ is processed through a dense layer for classification. This dense layer has weights $\mathbf{W} \in \mathbb{R}^{d_f \times N_C}$ and biases $\mathbf{b} \in \mathbb{R}^{N_C}$, where d_f is the dimension of the feature vector and N_C is the number of classes. The output of this dense layer is computed as:

$$\mathbf{z} = \mathbf{F}\mathbf{W} + \mathbf{b}, \quad (23)$$

where $\mathbf{z} \in \mathbb{R}^{N_C}$ is the logits vector.

The output layer applies the softmax activation function, defined as:

$$\sigma(\mathbf{z})_j = \frac{e^{z_j}}{\sum_{k=1}^{N_C} e^{z_k}}, \quad j = 1, \dots, N_C. \quad (24)$$

which ensures that the output probabilities sum to 1, enabling multi-class classification. The softmax function transforms the logits into a probability distribution over the N_C classes.

The final model output, $\hat{y}$, representing the predicted probability distribution, is given as:

$$\hat{y} = \sigma(\mathbf{z}). \quad (25)$$

This pipeline allows the network to map the feature vector $\mathbf{F}$ to a probability distribution over the target classes, thereby facilitating the prediction process.

F. Loss Function

The model is trained using the categorical cross-entropy loss, which measures the difference between the true distribution y and the predicted distribution $\hat{y}$. The loss is defined as:

$$\mathcal{L}(y, \hat{y}) = - \sum_{j=1}^{N_C} y_j \log(\hat{y}_j), \quad (26)$$

where $y_j \in \{0, 1\}$ indicates the true label for class j , and $\hat{y}_j \in [0, 1]$ is the predicted probability for class j . This loss function is well-suited for multi-class classification tasks as it penalizes incorrect predictions proportionally to their confidence levels.

VI. RESULTS

VII. EXPERIMENTAL SETUP

In this work, we utilize the datasets presented in [7], following the same experimental setup. We now provide a brief description of how these datasets were collected. To extract and evaluate the system's performance, we utilized the Nexmon CSI extraction tool and conducted experiments in diverse environments. The tool enables the acquisition of CSI data from Broadcom/Cypress Wi-Fi chipsets, such as the one found in the Asus RT-AC86U IEEE 802.11ac router. Operating in very-high-throughput mode, the system utilizes a total bandwidth of 80 MHz, with each CSI sample providing complex-valued channel information across 242 data sub-channels per transmit-receive antenna pair. For our configuration, with one transmitter antenna and four monitoring antennas, each CSI sample consists of four vectors containing 242 complex values each.

We evaluated the system under various conditions by conducting experiments in three distinct environments: a bedroom, a living room, and a university laboratory. Data was

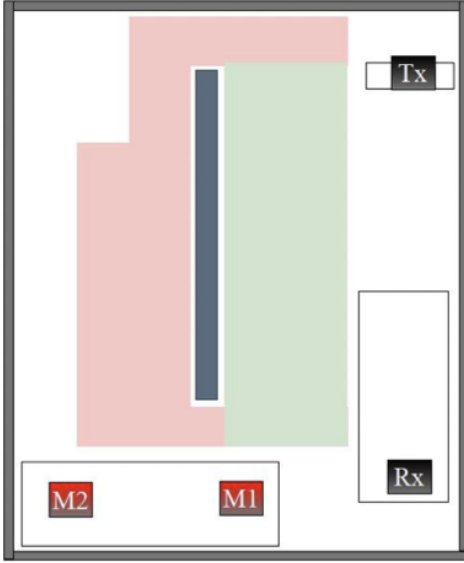


Fig. 7: Monitored environment for sets S1-S6: bedroom layout with occlusion caused by a bookcase. Tx and Rx denote the transmitter and receiver, respectively. Mj indicates the monitor station positions.

collected while volunteers performed activities such as walking, running, jumping, or sitting. The monitored environments and configurations are depicted in Figures 7 and 8.

Our setup included a transmitter (Tx), a receiver (Rx), and a monitor device configured in monitor mode, reflecting a real-world scenario where Wi-Fi traffic is present. The monitor node, implemented on an Asus router equipped with four antennas and Nexmon firmware, passively captured over-the-air packets to estimate the CFR. Signals of opportunity were generated using two Netgear X4S AC2600 Wi-Fi routers as the transmitter and receiver. Packets were transmitted at a fixed modulation and coding scheme (MSC 4) at a rate of 173 packets per second. This resulted in channel estimates every 6 milliseconds.

Each measurement campaign involved 120 seconds of CSI data for each activity, collected over different days and scenarios to ensure diversity in room geometry, environmental conditions, and participants. Table 1 summarizes the configurations, including positions of monitoring stations, activity areas, and the direct-path conditions between Tx and the monitor node. These datasets encompass over 120 minutes of CSI data, offering rich temporal and spatial diversity for training and validation.

VIII. NUMERICAL RESULTS

A. preprocessing step

The parameters used to train and test the MATC-Net model are presented in Tables 3. Table 4 summarizes the temporal efficiency metrics of the training and testing phases, in Table 5 report the performance of the model across the validation set and the ablation study, while in Table 6 reports the model's performance across different datasets.

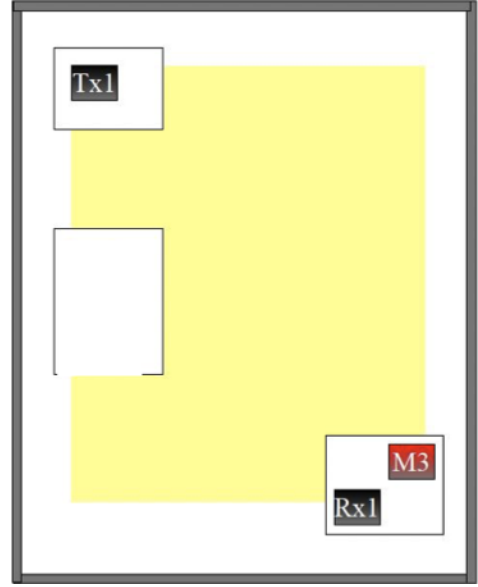


Fig. 8: Monitored environment for sets S7-S9: university laboratory setup with complex arrangements. Tx and Rx denote the transmitter and receiver, respectively. M4 indicates the monitor station.

TABLE 1: Measurement Conditions

Set	Monitor Position (M_j)	Direct Path	Type	Environment
S1	M2	yes	train-test	Bedroom
S2	M1	no	test	Bedroom
S3	M1	no	test	Bedroom
S4	M1	no	test	Bedroom
S5	M2	yes	test	Living Room
S6	M2	yes	test	Kitchen
S7	M3	no	test	Laboratory
S8	M3	no	test	Laboratory
S9	M3	no	test	Laboratory

The input signal consists of 242 subcarriers captured across 4 monitoring antennas. To increase the dataset size, the signal was segmented into window W of 200 samples with a 70% overlapping factor. A wavelet denoising step was applied during preprocessing using the 'sym4' wavelet transform and eventually a normalization is performed. Each segment is labeled with one of eight activity classes: Walking, Running, Sitting still, Standing still, Sitting down/Standing up, Empty, Jumping, and Doing arm exercises. The labels were encoded using one-hot encoding to match the requirements of the model. The entire model was trained and tested on a 2 GHz Intel Core i5 quad-core CPU.

B. Performance Analysis and Ablation Study

The dataset used for training and evaluation is S1 divided into training and validation sets in an 80:20 ratio. The performance of the MATC-Net model was evaluated on its ability to classify activities accurately across the combined dataset. As shown in Table 5 and in Figure 9, the system achieves high precision on the validation set with a reasonable temporal complexity. To further analyze the contributions of individual components of the model, we performed an ablation study

by isolating the performance of the time encoder tower and the channel encoder tower. As presented in Table 5, the time encoder achieves an average precision of 89.98% , while the channel encoder achieves 28.00%. These results highlight the complementary roles of the two towers in the model’s overall performance, with a dominant role performed by the time encoder.



Fig. 9: Confusion Matrix of the Validation Set

TABLE 2: Summary of Parameters Related to the Communication Setup

Parameter	Value
Monitored channel	IEEE 802.11ac
OFDM sample duration, T	3.2×10^{-6} s
Modulation and coding scheme (MCS)	4
Channel estimates interval, T_c	6×10^{-3} s
Number of OFDM sub-channels, M	256 (242 used)
Number of monitoring antennas, N_{ant}	4

In Figure 10, we present the PCA visualization of the feature fusion vectors derived from the validation test set. Each color corresponds to a specific activity, allowing us to observe the distribution of features across different activities. It is evident that the features tend to form distinct clusters by color, albeit not perfectly separated and with some degree of overlap.

This result is significant as it demonstrates how the transformer encoder effectively extracts meaningful features from temporal sequences. Furthermore, it highlights how these features are distributed differently depending on the performed activity. This provides valuable insights into the model’s capability to capture the underlying patterns of human activity through feature representation.

TABLE 3: Hyperparameters Used in the Learning Architectures

Hyperparameter	Value
Batch size	32
Number of classes	8
Sequence length	200
Overlapping samples	40
Antenna dropout, N_{drop}	0
Embedding dimension d_{emb}	128
Key dimension d_k	64
Number of layers	3
Number of heads	8
Dropout rate	0.1
Number of layers	3
Number of heads	8
Dropout rate	0.1
Optimizer	Adam
Learning rate	1×10^{-4}
Loss function	Categorical Crossentropy
Number of epochs	30
Training/Validation split	80/20

TABLE 4: Dataset and Training Details

Training set size	2615
Validation set size	654
Shape of data sample	200 x 968
Number of epochs	30
Training time	245 seconds
Avg. time for epoch	8.15 seconds
Avg. inference time for sample	0.009 seconds
Avg. re-training time	64 seconds

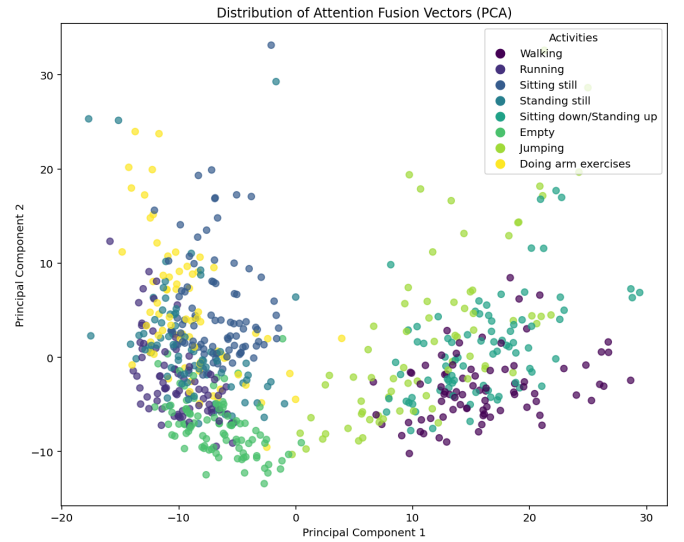


Fig. 10: Distribution of Attention Fusion Vectors

TABLE 5: MACT-Net Performance - Full Model

	EMP	SIT	WLK	RUN	JMP	STS	EXE	DSU	mean
Full Model	1.00	1.00	0.99	1.00	1.00	0.96	0.96	0.99	0.99
accuracy (%)	100	100	99	100	100	96.0	96.0	99.0	99.0
F1-score	1.00	1.00	0.99	1.00	1.00	0.96	0.96	0.99	0.99
Time Encoder Only	89.75	91.32	90.90	89.50	89.90	89.1	88.5	89.9	89.98
accuracy (%)	89.75	91.32	90.90	89.50	89.90	89.1	88.5	89.9	89.98
F1-score	0.897	0.913	0.909	0.895	0.899	0.891	0.885	0.899	0.899
Channel Encoder Only	0.31	0.39	0.30	0.35	0.28	0.24	0.12	0.30	0.28
accuracy (%)	31	39	30	35	28	24.0	12.0	30.0	28.0
F1-score	0.31	0.39	0.30	0.35	0.28	0.24	0.12	0.30	0.28

The acronyms used for activities are as follows: EMP = Empty, SIT = Sitting, WLK = Walking, RUN = Running, JMP = Jumping, STS = Standing still, EXE = Doing arm exercises, DSU = Sitting down/Standing up. This table reports the performance (accuracy and F1-score) of the full model, time-only encoder, and channel-only encoder on the validation set derived from the S1 training dataset.

C. Testing in New Environments and Transfer Learning

When applying the model to datasets collected in different environments, the classification performance was unfortunately suboptimal. To address this issue, we developed a transfer learning mechanism to refine the model weights for the target datasets. The objective was to use as few samples as possible for fine-tuning, simulating an initial calibration phase where the user performs a few test movements to adapt the system to the new environment. Specifically, we used only 10% of the new dataset as the re-training set and the remaining 90% as the test set. This transfer learning strategy allows the model to better generalize across varying environments by leveraging minimal re-training data. On average, the retraining time was approximately one minute, as shown in Table 4. Table 6 summarizes the results of the transfer learning experiments, showing improvements in classification performance across different test datasets.

IX. CONCLUDING REMARKS

In this paper, we have investigated a dual-transformer encoder architecture (MATC-Net) for WiFi-based Human Activity Recognition (HAR), focusing on both temporal and channel dimensions of CSI data. Our proposed approach effectively captures long-range dependencies and leverages a fusion mechanism to integrate temporal and channel-specific features. Through an ablation study, we have shown that each encoder tower contributes differently to the final recognition accuracy, with the temporal encoder playing a dominant role, and the channel encoder adding complementary information.

Beyond demonstrating robust performance on a controlled dataset, our results highlight that achieving reliable accuracy in new and dynamic environments remains a challenge. The transfer learning strategy we employed using minimal new training samples to adapt the model suggests that our approach is applicable to real world scenarios where continuous data collection might be costly or impractical. This adaptation mechanism is particularly relevant for practical HAR deployments, such as in healthcare monitoring or smart-home automation, where diverse environments and user behaviors require an agile model that can be quickly recalibrated.

TABLE 6: MACT-Net Performance on Test Datasets (S2–S10)

Dataset	Metric	EMP	SIT	WLK	RUN	JMP	STS	EXE	DSU	Mean
S2	Accuracy (%)	0.00	44.00	1.00	0.00	11.00	42.00	60.00	12.00	22.00
	F1-score	0.00	43.00	2.00	0.00	14.00	32.00	43.00	14.00	18.00
	Accuracy Retrain (%)	96.00	77.00	93.00	100.0	74.00	73.00	95.00	85.00	87.00
	F1 Retrain	98.00	77.00	90.00	100.0	77.00	74.00	90.00	87.00	87.00
S3	Accuracy (%)	60.00	33.00	3.00	0.00	0.00	16.00	14.00	28.00	20.00
	F1-score	64.00	23.00	5.00	0.00	0.00	14.00	12.00	32.00	19.00
	Accuracy Retrain (%)	94.00	86.00	95.00	94.00	100.0	71.00	95.00	95.00	91.00
	F1 Retrain	96.00	85.00	95.00	91.00	100.0	76.00	91.00	95.00	91.00
S4	Accuracy (%)	0.00	0.00	0.00	99.00	4.00	47.00	38.00	42.00	29.00
	F1-score	0.00	0.00	0.00	0.88	0.07	0.37	0.24	0.42	25.00
	Accuracy Retrain (%)	98.00	83.00	86.00	100.0	80.00	66.00	83.00	94.00	86.00
	F1 Retrain	0.98	0.74	0.84	1.00	0.86	0.72	0.88	0.89	86.00
S5	Accuracy (%)	0.00	59.00	22.00	1.00	89.00	23.00	52.00	33.00	35.00
	F1-score	0.00	40.00	34.00	2.00	68.00	23.00	41.00	43.00	31.00
	Accuracy Retrain (%)	97.00	88.00	87.00	91.00	86.00	71.00	91.00	76.00	86.00
	F1 Retrain	0.96	0.86	0.86	0.87	0.84	0.75	0.92	0.79	86.00
S6	Accuracy (%)	33.00	42.00	30.00	33.00	42.00	–	–	–	38.00
	F1-score	40.00	43.00	29.00	40.00	41.00	–	–	–	41.00
	Accuracy Retrain (%)	97.00	80.00	69.00	95.00	75.00	–	–	–	83.00
	F1 Retrain	0.96	0.80	0.72	0.93	0.73	–	–	–	83.00
S7	Accuracy (%)	4.00	3.00	30.00	1.00	0.00	41.00	6.00	0.00	11.00
	F1-score	6.00	4.00	36.00	1.00	1.00	24.00	4.00	4.00	10.00
	Accuracy Retrain (%)	94.00	55.00	66.00	92.00	78.00	51.00	75.00	79.00	74.00
	F1 Retrain	0.92	0.55	0.65	0.86	0.79	0.55	0.77	0.78	74.00
S8	Accuracy (%)	1.00	17.00	–	6.00	–	36.00	29.00	–	19.00
	F1-score	2.00	20.00	–	9.00	–	35.00	28.00	–	19.00
	Accuracy Retrain (%)	88.00	66.00	–	96.00	–	78.00	78.00	–	82.00
	F1 Retrain	0.87	0.73	–	0.88	–	0.79	0.79	–	82.00
S9	Accuracy (%)	1.00	7.00	–	1.00	–	22.00	18.00	–	10.00
	F1-score	1.00	8.00	–	1.00	–	21.00	14.00	–	10.00
	Accuracy Retrain (%)	96.00	71.00	–	87.00	–	80.00	84.00	–	86.00
	F1 Retrain	0.90	0.76	–	0.93	–	0.82	0.84	–	86.00

Cells marked with '–' indicate activities not present in a given dataset.

Nevertheless, several open issues remain. First, more extensive cross-domain validation is needed to further confirm the model's generalization in diverse settings with different device types and network protocols. Second, while we used minimal data for transfer learning, future extensions could investigate few-shot or meta-learning frameworks to reduce the re-training overhead even further. Additionally, online adaptation methods could be explored to continuously refine the model in real-time.

From a project standpoint, we have learned the intricacies of exploiting transformer-based architectures for time-series data and the importance of carefully managing high-dimensional CSI inputs. Balancing computation time with model complexity proved critical, particularly as we integrated multiscale convolutions and attention fusion. We also encountered some difficulties, mainly regarding data acquisition synchronization and noise filtering: real-world WiFi channels introduce variability that requires robust preprocessing and calibration. Ensuring reproducible training conditions across different hardware setups was another non trivial challenge, underscoring the need for standardized testing environments.

In summary, our proposed MATC-Net architecture provides an effective way to combine temporal and channel features for HAR, with promising real-world applicability. By extending transfer learning strategies and addressing the challenges listed

above, we believe this framework can be further refined to handle a broader range of environments and activity types in the future.

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